

Manufacturing Process Health and Operational Efficiency Analysis in 6G-Enabled Smart Factories

Background and Context:

Modern smart factories operate with:

- Hundreds of interconnected machines
- Real-time Industrial IoT (IIoT) sensors
- High dependency on stable production quality

Before applying AI for prediction or optimization, manufacturing leaders must first answer:

- Are machines operating within healthy ranges ?
- Where are efficiency losses occurring ?
- Which machines or operation modes show early signs of degradation ?



Problem Statement

Despite having large volumes of real-time sensor and production data, manufacturing teams often struggle with:

- Lack of centralized visibility into machine health
- Difficulty correlating sensor behavior with production efficiency
- Reactive handling of defects and errors
- Inability to benchmark machines against operational standards



As a result, inefficiencies go unnoticed until they impact production output or quality.

Dataset Fields Description:

Column Name	Description
Date	Calendar date on which the machine data was recorded.
Time	Timestamp indicating the exact time of data capture.
Machine_ID	Unique identifier assigned to each industrial machine.
Operation_Mode	Current operating mode of the machine (e.g., normal, high-load).
Temperature_C	Operating temperature of the machine in degrees Celsius.
Vibration_Hz	Machine vibration frequency measured in Hertz
Power_Consumption_kW	Electrical power consumed by the machine in kilowatts.
Network_Latency_ms	Communication latency experienced over the 6G network in milliseconds.
Packet_Loss_%	Percentage of data packets lost during machine communication.
Quality_Control_Defect_Rate_%	Percentage of defective units identified during quality checks.
Production_Speed_units_per_hr	Number of units produced per hour by the machine.
Predictive_Maintenance_Score	AI-derived score indicating the machine's maintenance readiness.
Error_Rate_%	Percentage of operational errors occurring during production.
Efficiency_Status	Target label classifying manufacturing efficiency as High, Medium, or Low.

Analytical Methodology (Step-by-Step)

Data Validation & Preparation

- Validate sensor ranges and units
- Handle missing or inconsistent time records
- Standardize machine identifiers and operation modes

Machine-Level Sensor Health Analysis

- Analyze temperature, vibration, and power patterns per machine
- Identify machines operating near threshold limits
- Compare sensor stability across operation modes

Production Performance Diagnostics

- Analyze production speed trends
- Compare output consistency across machines
- Identify underperforming machines

Quality & Error Analysis

- Correlate defect rate with sensor behavior
- Analyze error rate spikes by machine and time
- Identify quality bottlenecks

Efficiency Status Distribution

- Evaluate how often machines fall into:
 - High efficiency
 - Medium efficiency
 - Low efficiency
- Compare efficiency across operation modes and shifts

Cross-Metric Diagnostics

- Temperature vs defect rate analysis
- Vibration vs error rate relationship
- Power consumption vs efficiency trade-offs

Key Performance Indicators (KPIs)

KPI	Description
Machine Health Index	Composite score from temperature, vibration, power
Average Production Speed	EMean output rate per machine
Defect Density Score	Defect rate relative to production volume
Error Frequency Index	Rate of operational errors
Efficiency Distribution	High / Medium / Low efficiency spread

Streamlit Web Application Requirements

Dashboard Modules

Factory Health Overview

- Overall efficiency distribution
- Average sensor metrics

Machine Health Dashboard

- Machine-wise sensor trends
- Health scorecards

Production & Quality Panel

- Production speed vs defect rate
- Error frequency visualizations

Efficiency Diagnostics View

- Efficiency_Status breakdown, Machine and mode comparisons

User Capabilities

- Machine selector
- Operation mode filter
- Date & time range selector
- Metric comparison toggles

Deliverables and Submission

- Research paper (EDA, insights, recommendations)
- Streamlit dashboard (live analytics)
- Executive summary for government stakeholders

Conclusion

This project establishes a diagnostic intelligence layer for 6G-enabled smart manufacturing environments. By systematically analyzing machine health, production efficiency, and quality indicators, it empowers Thales and its industrial partners to move from reactive issue handling to data-driven operational excellence. This project forms the critical first step before predictive maintenance, efficiency forecasting, or AI-driven optimization initiatives

